

Mark Jennings

Robotics Engineer · markjennings97@gmail.com · makr.org

Summary

Robotics engineer with 7+ years of experience developing integrated robotic hardware from concept through deployment. Hands-on background in mechanical design, electro-mechanical actuation, software development, system integration/validation, camera calibration, and field deployment. Relocating to Denver in December 2026.

Experience

Appttronik

July 2026 - Present

Senior Hardware Integration Engineer

- Own bring-up and validation of next-generation humanoid head/neck hardware, integrating mechanical, electrical, and software interfaces and performing sub-system endurance testing.
- De-risk long-horizon behaviors (e.g. autonomous battery swapping) with representative hardware prior to full system bring-up.
- Establish in-house calibration capabilities to improve repeatability and characterization of custom perception stack.

Elevate Robotics (Appttronik spin-off)

September 2024 - June 2026

Robotics Engineer

- Designed a lightweight, high-payload robotic manipulator with an industry-leading >1:1 payload-to-weight ratio from concept to an upcoming product, driving mechanical architecture, structural analysis, actuation, packaging, manufacturability, safety compliance, and cost optimization.
- Led technical analysis for a NASA SBIR effort and secured \$1.275M in Phase II funding by translating robotic system capabilities and development plans into a compelling technical proposal.
- Deployed prototype robotic hardware to construction sites and trade shows, supporting field integration, debugging, and feedback into subsequent design iterations.
- Initiated 4 patents for concepts ranging from industrial design to a novel fail-safe braking system.

Los Alamos National Laboratory

October 2021 - August 2024

R&D Engineer

- Overhauled a nuclear manufacturing glovebox with the first autonomous robotic arm used in U.S. plutonium part production, integrating an industrial robot arm, custom end-effector, alignment sensors, and operator workflows in a highly regulated and safety critical environment.
- Created control software, operating procedures, and custom tooling for multiple nuclear manufacturing processes, taking systems from conceptual ideas to daily operational use.
- Coordinated multi-year intern program to deploy heterogeneous robot fleet for mock radiation survey.

Nuclear and Applied Robotics Group

August 2019 - August 2021

Graduate Research Assistant

- Researched physical human-robot interaction, augmenting human dexterity/reasoning with robotic precision/strength.
- Developed C++ packages for gravity-compensated manipulator (precursor to Elevate Arm), custom mobile manipulator, and perception calibration routines.

Skills

- **Mechanical Design:** CAD (Onshape, SolidWorks, NX, Creo), GD&T, DFMA, FEA, linkage/mechanism optimization, stress analysis, industrial design, human factors
- **Fabrication:** CNC/manual machining, additive manufacturing, casting, injection molding, extruding, sheet metal bending, welding, rapid prototyping, general shop and hand tool operation
- **Electric Actuation:** Rotor/stator selection, motor drivers, linear/rotary actuation, reduction (planetary and strain wave), brakes, bearings, encoders, controls
- **Software:** C/C++, Python, Linux, Git, ROS/ROS2, physics simulation, manipulator motion planning, mobile robot localization/navigation/mapping, camera calibration, sensor fusion
- **Testing/Validation:** Testbed design, instrumentation, endurance/proof/destructive testing, failure analysis

Education

The University of Texas at Austin

Master of Science in Mechanical Engineering (3.96)

August 2021

Bachelor of Science in Mechanical Engineering (3.84)

May 2019